

ASSESSMENT TOOL – 2

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO2	Assessment	Tool – Scenario Based Qns
Weightage		Total Marks	

Student Name	Cindy R	Reg. No.	192412571
Section / Batch	ECE-2024	Date of Submission	29/7/26
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks	20	Obtained Marks	20
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Code of Conduct:

I Cindy R / 192412571 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Scenario 1

Given Network

Network Block 192.168.100.0/24

Department	Prefix
School of Computing	/25
School of Electronics	/26
School of Mechanical Engineering	/27

Step 1: Subnet Allocation

School of Computing (/25)

Subnet Mask:

255.255.255.128

Network Address:

192.168.100.0

Broadcast Address:

192.168.100.127

Host Range:

192.168.100.1 – 192.168.100.126

Usable Hosts:

$$2^7 - 2 = 126$$

School of Electronics (/26)

Subnet Mask:

255.255.255.192

Network Address:

192.168.100.128

Broadcast Address:

192.168.100.191

Host Range:

192.168.100.129 – 192.168.100.190

Usable Hosts:

$$2^6 - 2 = 62$$

School of Mechanical Engineering (1/27)

Subnet Mask:

255.255.255.224

Network Address:

192.168.100.192

Broadcast Address:

192.168.100.223

Host Range:

192.168.100.193 – 192.168.100.222

Usable Hosts:

$$2^5 - 2 = 30$$

Remaining Unused IP Range

Unused addresses:

192.168.100.224 – 192.168.100.255

Total remaining addresses:

32

Summary

Department	Network	Host Range	Broadcast	Usable Hosts
Computing	192.168.100.0/25	1–126	127	126
Electronics	192.168.100.128/26	129–190	191	62
Mechanical	192.168.100.192/27	193–222	223	30

Unused:

192.168.100.224–255



Scenario 1

Given Network

10.10.0.0/16

Department	Prefix
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

Software Development (/24)

Network

10.10.0.0

Broadcast

10.10.0.255

Host Range

10.10.0.1 - 10.10.0.254

Usable Hosts

$$2^8 - 2 = 254$$

Quality Assurance (/25)

Network

10.10.1.0

Broadcast

10.10.1.127

Host Range

10.10.1.1 - 10.10.1.126

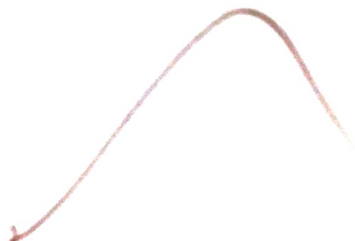
Usable Hosts

126

Human Resources (/26)

Network

10.10.1.128



Broadcast:
10.10.1.191

Host Range:
10.10.1.129–10.10.1.190

Usable Hosts:
62

Executive Management (/27)

Network:
10.10.1.192

Broadcast:
10.10.1.223

Host Range:
10.10.1.193–10.10.1.222

Usable Hosts:
30

Remaining Unused Range

Allocated:

10.10.0.0 – 10.10.1.223

Remaining:

10.10.1.224 – 10.10.255.255

Summary

Department	Network	Host Range	Broadcast	Hosts
Software Development	10.10.0.0/24	1–254	255	254
Quality Assurance	10.10.1.0/25	1–126	127	126
Human Resources	10.10.1.128/26	129–190	191	62
Executive	10.10.1.192/27	193–222	223	30

Unused:

10.10.1.224–10.10.255.255

Scenario 3

Given IPv6

2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Form

2001:db8:abcd::/64

2. Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix

/64 means first 64 bits represent the network.

Network Prefix:

2001:db8:abcd:0000::/64

4. Number of Host Addresses

Host bits

$$128 - 64 = 64 \text{ bits}$$

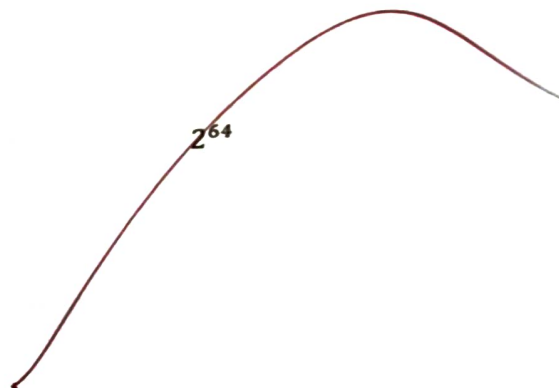
Therefore

Host addresses

$$= 18,446,744,073,709,551,616$$

5. Total Possible Hosts

$$18,446,744,073,709,551,616$$



Summary

Item	Answer
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Network Prefix	2001:db8:abcd:0000::/64
Host Addresses	2^{64}
Total Hosts	18,446,744,073,709,551,616

Scenario 4

Routing Table

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP

192.168.2.45

1. Destination Subnet

192.168.2.0/24

2. Matching Entry

192.168.2.0/24

3. Next-Hop Interface

Forward through the interface connected to

192.168.2.0/24

4. Default Route

If destination is not found,

Use

0.0.0.0/0

(Default Gateway)

Summary

Item	Answer
Destination Subnet	192.168.2.0/24
Routing Entry	192.168.2.0/24
Forward Interface	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0

Scenario 5

Administration LAN

192.168.10.0/26

Finance LAN

192.168.10.64/26

Gateways

Administration:

192.168.10.1

Finance:

192.168.10.65

Administration LAN

Network Address

192.168.10.0

Broadcast

192.168.10.63

Host Range

192.168.10.1–192.168.10.62

Suggested PCs

PC1

192.168.10.2

PC2

192.168.10.3

PC3

192.168.10.4

Gateway

192.168.10.1

Finance LAN

Network

192.168.10.64

Broadcast